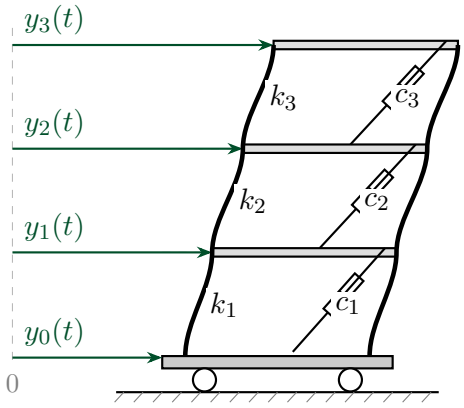
1

MA3521 — Three-Story Building

Name:

Displacements: $\mathbf{y}(t) = (y_1(t), y_2(t), y_3(t))$ (m), t in seconds.



$m_3 = 2\,\text{kg}$
$k_3 = 2\,\text{N/m}$
$c_3 = 1\,\text{N s/m}$

$m_2 = 3\,\text{kg}$
$k_2 = 3\,\text{N/m}$
$c_2 = 1\,\text{N s/m}$

$m_1 = 4\,\text{kg}$
$k_1 = 4\,\text{N/m}$
$c_1 = 2\,\text{N s/m}$

$y_0(t) = \sin(t)$ m

Initial displacements: $\mathbf{y}(0) = (0, -1, 0)$ m.
Of course, $\mathbf{y}'(t) = (y_1'(t), y_2'(t), y_3'(t))$.
Initial velocities: $\mathbf{y}'(0) = (0, 0, 2)$ m/s.

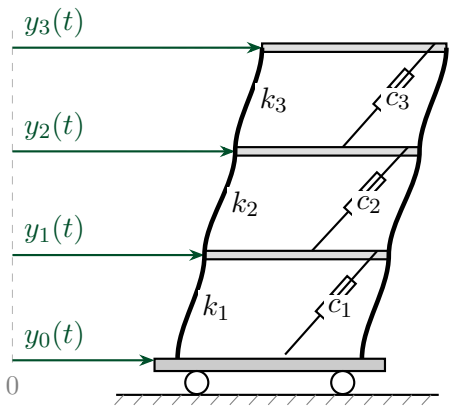
1a. Fill in $M\mathbf{y}'' + C\mathbf{y}' + K\mathbf{y} = \mathbf{b}(t)$. Do not solve.

$$M = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \quad C = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \quad K = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \quad \mathbf{b}(t) = \begin{pmatrix} \\ \\ \end{pmatrix}$$

1b. With $\mathbf{z} = (\mathbf{y}, \mathbf{y}') = (y_1, y_2, y_3, y_1', y_2', y_3')$, fill in $\mathbf{z}' = A\mathbf{z} + \mathbf{g}(t)$ and $\mathbf{z}(0) = \mathbf{z}_0$.

$$A = \begin{pmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{pmatrix} \quad \mathbf{g}(t) = \begin{pmatrix} \\ \\ \\ \\ \\ \end{pmatrix} \quad \mathbf{z}_0 = \begin{pmatrix} \\ \\ \\ \\ \\ \end{pmatrix}$$

Displacements: $\mathbf{y}(t) = (y_1(t), y_2(t), y_3(t))$ (m), t in seconds.



$m_3 = 1\,\text{kg}$
$k_3 = 2\,\text{N/m}$
$c_3 = 1\,\text{N s/m}$

$m_2 = 2\,\text{kg}$
$k_2 = 4\,\text{N/m}$
$c_2 = 2\,\text{N s/m}$

$m_1 = 3\,\text{kg}$
$k_1 = 6\,\text{N/m}$
$c_1 = 3\,\text{N s/m}$

$y_0(t) = 2\sin(t)$ m

Initial displacements: $\mathbf{y}(0) = (0, 0, 1)$ m.
Of course, $\mathbf{y}'(t) = (y_1'(t), y_2'(t), y_3'(t))$.
Initial velocities: $\mathbf{y}'(0) = (-1, 0, 0)$ m/s.

2a. Fill in $M\mathbf{y}'' + C\mathbf{y}' + K\mathbf{y} = \mathbf{b}(t)$. Do not solve.

$$M = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \quad C = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \quad K = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \quad \mathbf{b}(t) = \begin{pmatrix} \\ \\ \\ \end{pmatrix}$$

2b. With $\mathbf{z} = (\mathbf{y}, \mathbf{y}') = (y_1, y_2, y_3, y_1', y_2', y_3')$, fill in $\mathbf{z}' = A\mathbf{z} + \mathbf{g}(t)$ and $\mathbf{z}(0) = \mathbf{z}_0$.

$$A = \begin{pmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{pmatrix} \quad \mathbf{g}(t) = \begin{pmatrix} \\ \\ \\ \\ \\ \\ \end{pmatrix} \quad \mathbf{z}_0 = \begin{pmatrix} \\ \\ \\ \\ \\ \\ \end{pmatrix}$$